# SAMPANNA SHETTY

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## SUMMARY

Artificial Intelligence and Machine Learning undergraduate with research experience in Computer Vision, Deep Learning, NLP, and Optimization Algorithms. Experienced in developing AI-driven applications, conducting applied research, and building scalable solutions for real-world challenges.

## EDUCATION

**Bachelor of Technology (B.Tech) in Artificial Intelligence & Machine Learning** 2023–2027 NMAM Institute of Technology, Nitte
CGPA: 8.97/10

## SKILLS

**Programming Languages:** Python, C++, SQL

**Artificial Intelligence & Machine Learning:** PyTorch, Scikit-Learn, OpenCV, YOLO, Deep Learning, Machine Learning, NLP, Neural Networks, Prompt Engineering

**Data Science & Analytics:** Pandas, NumPy, Feature Engineering, Predictive Modeling, Data Analysis

**Frameworks & Tools:** Flask, Git, Linux, Arduino, CuPy, GPU Acceleration

**Web Technologies:** Angular, React

**Coursework:** Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks

## PUBLICATIONS

**GPU Framework for Priority-Aware Dynamic Routing Optimization** 2025
Co-Author & Presenter

- Engineered a GPU-accelerated Genetic Algorithm and Simulated Annealing framework across **3 optimization strategies**, achieving **2× faster optimization** than CPU-based implementations with **60% lower compute overhead**.
- Enhanced route optimization quality by **25%** and reduced convergence time by **40%** across datasets containing **2,000+ nodes** and **500+ routing constraints**.
- Scaled routing computations to support **10+ large-scale** emergency response and logistics scenarios, improving resource utilization by **35%** across **4 benchmark datasets**.

**Assistive Vision for Visually Impaired: An Integrated Deep Learning and IoT Framework for Real-Time Object Recognition and Navigational Assistance** 2026
Co-Author

- Built a YOLOv5-based assistive vision framework detecting and classifying **20+ everyday object categories** across **3 lighting conditions** in real time at **30+ FPS**.
- Integrated smartphone-based computer vision with **2 ESP32-enabled IoT modules**, enabling **3-mode** multimodal audio and alert-based navigation assistance with **under 200ms latency**.
- Achieved **96.2% accuracy**, **95.8% precision**, **94.6% recall**, and **96.8% mAP@0.5**, outperforming **5+ baseline** object detection models including SSD and MobileNet.

**Intelligent Rain Sensing Wiper Technology** 2026
Co-Author & Presenter

- Developed an Arduino-based intelligent rain-sensing wiper system with automated speed adjustment across **3 rainfall intensity levels**, validated under **200+ test conditions**.
- Achieved **96–98% rainfall detection accuracy** with response times as low as **0.75 seconds**, outperforming **2 conventional wiper control baselines**.
- Improved driving visibility by **40%** through real-time environmental sensing and adaptive control mechanisms across **4 simulated weather scenarios**.

## PROJECTS

**Emotion Detection System**

- Built a real-time emotion recognition system using OpenCV, Flask, and **3 ML models**, processing **25+ frames per second** with end-to-end inference latency under **100ms**.

- Classified **7 emotional states** with over **90% accuracy** on **5,000+ facial images** across **4 demographic groups**, reducing misclassification by **18%** over baseline CNN.

**Neonatal Sepsis Prediction**

- Developed a Random Forest model for early-onset neonatal sepsis prediction using **15+ clinical features** extracted from NICU datasets, achieving **89% AUC-ROC**.
- Trained and evaluated across **1,500+ patient records** through **5-fold cross-validation** and feature engineering, reducing false negatives by **22%** over logistic regression baseline.

**Story Generation using NLP**

- Built a Transformer-based NLP model for context-aware story generation using **4-layer architecture** with **8 attention heads** and **512-dimensional** embeddings.
- Trained on **10,000+ text samples** generating coherent **5+ paragraph** stories, achieving a **BLEU score of 0.72** across **3 narrative genres**.

**Vajrayudh – Real-Time Monitoring System**

- Architected a computer vision-based monitoring system across **4 detection modules** for automated event detection, processing **1080p video streams** at **24 FPS**.
- Processed **500+ monitoring events** with **93% detection precision**, reducing manual surveillance effort by **70%** and alert response time by **55%**.

## CERTIFICATIONS

- Google AI Essentials – Google
- Prompt Design in Vertex AI Skill Badge – Google Cloud
- Microsoft Learn AI Fundamentals
- Microsoft Applied AI Skills
- AWS Academy Cloud Foundations
- HP Data Science & Analytics
- Web Development (Basic) – HackerRank

## LEADERSHIP & ACTIVITIES

**Branch Captain – Department of Artificial Intelligence & Machine Learning** 2025–Present

- Represented **120+ AI & ML students** as liaison between the student body, **15+ faculty**, and administration, facilitating **10+ initiatives** and boosting participation by **30%**.

**Operations Manager – FiniteLoop Club** 2025–Present

- Directed operations for **100+ member** developer community, co-organizing **HackFest 2026** with **500+ participants** across **30+ teams** and **8+ technical workshops**.

**Secretary – NMAMIT Student Research Forum (NSRF)** 2025–Present

- Coordinated **100+ researchers** and **10+ mentors**, organizing a conference with **70+ paper presentations** across **4 tracks**, reaching **300+ students** institute-wide.

**Co-Head, Event Management Committee – Incridea'26** 2026

- Spearheaded **15+ events** over **3 days** for **5000+ attendees**, supervising **50+ volunteers** across **10+ zones** with **95%+ participant satisfaction**.